Solutions

Maths

Q1. Let k be a real constant defined by $k = \lim_{x \to 0} \frac{\cos(xe^{-x}) - \cos(xe^{x})}{2x^3}$. Consider two non-intersecting curves C_1 and C_2 in the xy-plane defined by the equations: $C_1 : y = e^x + 2k$ $C_2 : y = \ln(x - 2k)$ (for $x > 2k$). If d is the minimum distance between the curves C_1 and C_2, then d is equal to:

(A) $2\sqrt{2}$ (B) $3\sqrt{2}$ (C) $4\sqrt{2}$ (D) $5\sqrt{2}$

Ans. B

Sol. $$k = \lim_{x \to 0} \frac{\left(1 - \frac{x^2}{2} + x^3\right) - \left(1 - \frac{x^2}{2} - x^3\right) + O(x^4)}{2x^3}$$

$$= \lim_{x \to 0} \frac{2x^3 + O(x^4)}{2x^3} = 1$$

With $k = 1$, the curves are: $C_1 : y = e^x + 2$ and $C_2 : y = \ln(x - 2)$

Since C_1 and C_2 are inverse functions, they are symmetric with respect to the line of symmetry $y = x$.

The minimum distance d occurs along the common normal having slope -1.

For C_1, the point of closest approach is $P(x_1, y_1)$ where the tangent is parallel to $y = x$: $\frac{dy}{dx} = e^{x_1} = 1$

$\implies x_1 = 0 \implies P(0, 3)$

For C_2, the point of closest approach is $Q(x_2, y_2)$ where the tangent is parallel to $y = x$: $\frac{dy}{dx} = \frac{1}{x_2 - 2} = 1$

$\implies x_2 = 3 \implies Q(3, 0)$

The segment PQ has a slope of -1, which is perpendicular to both tangents.

Thus, the minimum distance d is: $d = \sqrt{(3-0)^2 + (0-3)^2} = 3\sqrt{2}$

Q2. If $[t]$ denotes the greatest integer less than or equal to t, then the value of the definite integral $I = \int_0^{\pi} ([2\sin 2x] + [2\cos 2x])dx$ is equal to:

(A) 0 (B) $-\frac{\pi}{2}$ (C) $-\pi$ (D) -2π

Ans. C

Sol. Split the integral into two parts: $I = I_1 + I_2$, where $I_1 = \int_0^{\pi} [2\sin 2x]dx$ & $I_2 = \int_0^{\pi} [2\cos 2x]dx$

Applying the Queen property $\int_0^{2a} f(x)dx = \int_0^{a} (f(x) + f(2a-x))dx$ to I_1 with $2a = \pi$:

$= \int_0^{\pi/2} ([2\sin 2x] + [2\sin(2(\pi - x))])dx = \int_0^{\pi/2} ([2\sin 2x] + [-2\sin 2x])dx$

$I_1 = \int_0^{\pi/2} (-1)dx = -\frac{\pi}{2}$

Similarly, applying the Queen property to I_2 with $2a = \pi$:

$I_2 = \int_0^{\pi/2} ([2\cos 2x] + [2\cos(2(\pi - x))])dx = 2\int_0^{\pi/2} [2\cos 2x]dx$

$= 2\int_0^{\pi/4} ([2\cos 2x] + [2\cos(2(\pi/2 - x))])dx = 2\int_0^{\pi/4} ([2\cos 2x] + [-2\cos 2x])dx$

Since $2\cos 2x$ is not an integer almost everywhere on $(0, \pi/4)$, the integrand simplifies to -1:

$I_2 = 2\int_0^{\pi/4} (-1)dx = -\frac{\pi}{2}$

$I = I_1 + I_2 = -\frac{\pi}{2} - \frac{\pi}{2} = -\pi$

Q3. Let $f : (1, \infty) \to \mathbb{R}$ be a function defined by

$$f(x) = \int \frac{x^2 + 1}{x^4 - x^2} dx$$

satisfying the condition $f(2) = \frac{1}{2} - \ln 3$. Consider the curves $C_1 : y = f(x)$ and $C_2 : y = \frac{1}{x} - \ln(x+1)$. If the acute angle of intersection between the curves C_1 and C_2 is θ, then θ is equal to

(A) $\tan^{-1}\left(\frac{144}{109}\right)$ (B) $\tan^{-1}\left(\frac{12}{7}\right)$ (C) $\tan^{-1}\left(\frac{7}{12}\right)$ (D) $\tan^{-1}\left(\frac{109}{144}\right)$

Ans. A

Sol. $\frac{x^2 + 1}{x^2(x^2 - 1)} = \frac{(x^2 - 1) + 2}{x^2(x^2 - 1)}$

$= \frac{1}{x^2} + \frac{2}{x^2 - 1} - \frac{2}{x^2} = \frac{2}{x^2 - 1} - \frac{1}{x^2}$

$f(x) = \ln\left(\frac{x-1}{x+1}\right) + \frac{1}{x} + C$

Using the initial condition $f(2) = \frac{1}{2} - \ln 3$: $\frac{1}{2} + \ln\left(\frac{1}{3}\right) + C = \frac{1}{2} - \ln 3 \implies C = 0$

Thus, the curve C_1 is given by: $y = \frac{1}{x} + \ln(x-1) - \ln(x+1)$

To find the point of intersection of C_1 and C_2, $\frac{1}{x} + \ln(x-1) - \ln(x+1)$

$= \frac{1}{x} - \ln(x+1) \Rightarrow \ln(x-1) = 0 \Rightarrow x = 2$

The slope of the tangent to C_1 at $x = 2$ is: $m_1 = f'(2) = \frac{2^2 + 1}{2^4 - 2^2} = \frac{5}{12}$

$$g'(x) = -\frac{1}{x^2} - \frac{1}{x+1} \Rightarrow m_2 = g'(2) = -\frac{1}{4} - \frac{1}{3} = -\frac{7}{12}$$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\frac{5}{12} - \left(-\frac{7}{12}\right)}{1 + \left(\frac{5}{12}\right)\left(-\frac{7}{12}\right)}\right| = \frac{1}{1 - \frac{35}{144}} = \frac{144}{109}$$

$$\theta = \tan^{-1}\left(\frac{144}{109}\right)$$

Q4. Let $L = \lim\limits_{n\to\infty} \sum\limits_{k=1}^{n-1} \dfrac{n^2}{\sqrt{n^6 - k^6 + n^2k^2(n^2-k^2)}}$. Then the value of L is

(A) $\dfrac{\pi}{4}$ (B) $\dfrac{\pi}{2\sqrt{2}}$ (C) $\dfrac{\pi}{\sqrt{2}}$ (D) $\dfrac{\pi}{2}$

Ans. B

Sol. $$T_k = \frac{n^2}{\sqrt{n^6 - k^6 + n^2k^2(n^2-k^2)}}$$

$$n^6 - k^6 + n^2k^2(n^2-k^2)$$

$$= (n^2-k^2)(n^4 + n^2k^2 + k^4) + n^2k^2(n^2-k^2)$$

$$= (n^2-k^2)(n^4 + 2n^2k^2 + k^4) = (n^2-k^2)(n^2+k^2)^2$$

$$T_k = \frac{n^2}{(n^2+k^2)\sqrt{n^2-k^2}}$$

$$= \frac{1}{n} \cdot \frac{1}{\left(1 + \left(\frac{k}{n}\right)^2\right)\sqrt{1 - \left(\frac{k}{n}\right)^2}}$$

$$L = \int_0^1 \frac{dx}{(1+x^2)\sqrt{1-x^2}}$$

$x = \sin\theta \implies dx = \cos\theta d\theta$:

$$L = \int_0^{\frac{\pi}{2}} \frac{\cos\theta d\theta}{(1+\sin^2\theta)\sqrt{1-\sin^2\theta}} = \int_0^{\frac{\pi}{2}} \frac{d\theta}{1+\sin^2\theta}$$

Dividing the numerator and denominator by $\cos^2\theta$:

$$L = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta d\theta}{\sec^2\theta + \tan^2\theta} = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta d\theta}{1 + 2\tan^2\theta}$$

Substituting $\tan\theta = t \implies \sec^2\theta d\theta = dt$:

$$L = \int_0^{\infty} \frac{dt}{1+2t^2} = \frac{1}{2}\int_0^{\infty} \frac{dt}{t^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$L = \frac{1}{2} \cdot \sqrt{2}\left[\tan^{-1}(\sqrt{2}t)\right]_0^{\infty} = \frac{1}{\sqrt{2}}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{2\sqrt{2}}$$

Q5. Let $f : (0,1) \to \mathbb{R}$ be a continuously differentiable function satisfying the differential relation $df(x) = \dfrac{dx}{x^2\sqrt{1-x^2}}$ for all $x \in (0,1)$. Suppose the curve $y = f(x)$ passes through the point $\left(\dfrac{1}{2}, -\sqrt{3}\right)$ and is denoted by C_1. Let C_2 be the curve defined by the equation $x^2 = \dfrac{3}{2}y$. If a

line L is a common tangent to both the curves C_1 and C_2, touching C_1 at the point P and C_2 at the point Q, then which of the following statements is/are **TRUE**?

(A) $f(x) = -\dfrac{\sqrt{1-x^2}}{x}$

(B) One possible equation of the common tangent line L is $2\sqrt{2}x - y - 3 = 0$

(C) The length of the line segment PQ is $3\sqrt{2}$

(D) The area of the triangle formed by the line L with the coordinate axes is $\dfrac{9}{2\sqrt{2}}$

Ans. A, B, C

Sol. Integrating $df(x) = \dfrac{dx}{x^2\sqrt{1-x^2}}$ using the substitution $x = \sin\theta$ yields $f(x) = -\cot\theta + C = -\dfrac{\sqrt{1-x^2}}{x} + C.$

Applying the initial condition $f\left(\dfrac{1}{2}\right) = -\sqrt{3}$ gives $C = 0$, so $f(x) = -\dfrac{\sqrt{1-x^2}}{x}$.

The slope of the tangent to C_1 at any point is $y' = \dfrac{1}{x^2\sqrt{1-x^2}}$, and for C_2, $y = \dfrac{2}{3}x^2 \implies y' = \dfrac{4}{3}x$.

Let the point of tangency on C_1 be $P(x_1, y_1)$.

Choosing $x_1 = \dfrac{1}{\sqrt{2}}$ gives the slope $m = 2\sqrt{2}$ and the point $P\left(\dfrac{1}{\sqrt{2}}, -1\right)$.

The equation of the tangent at P is $y - (-1) = 2\sqrt{2}\left(x - \dfrac{1}{\sqrt{2}}\right) \Rightarrow 2\sqrt{2}x - y - 3 = 0,$

For C_2, setting the slope $\dfrac{4}{3}x_2 = 2\sqrt{2}$ gives $x_2 = \dfrac{3\sqrt{2}}{2}$, which corresponds to the point $Q\left(\dfrac{3\sqrt{2}}{2}, 3\right)$.

Substituting Q into the line equation gives $2\sqrt{2}\left(\dfrac{3\sqrt{2}}{2}\right) - 3 - 3 = 0$, confirming it is a valid common tangent.

The distance $PQ = \sqrt{\left(\dfrac{3\sqrt{2}}{2} - \dfrac{1}{\sqrt{2}}\right)^2 + (3-(-1))^2}$

$= \sqrt{(\sqrt{2})^2 + 4^2} = 3\sqrt{2},$

The intercepts of the line $2\sqrt{2}x - y = 3$ are $\left(\dfrac{3}{2\sqrt{2}}, 0\right)$ and $(0, -3)$.

The area of the triangle formed with the coordinate axes is $\dfrac{1}{2} \cdot \dfrac{3}{2\sqrt{2}} \cdot |-3| = \dfrac{9}{4\sqrt{2}},$

Q6. Let $f : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function satisfying $f(0) = 0$, $f'(0) = 2$, and $f''(0) = -1$. Let $g : \mathbb{R} \to \mathbb{R}$ be a composite function defined by $g(x) = f(x + f(x))$. Which of the

following statements is (are) **TRUE**?

(A) $g'(0) = 6$ (B) $\lim\limits_{x\to 0} \dfrac{g(x) - 6x}{x^2} = -\dfrac{11}{2}$ (C) $\lim\limits_{x\to 0} \dfrac{f(g(x)) - 12x}{x^2} = -29$

(D) $\lim\limits_{x\to 0} \dfrac{g(f(x)) - 12x}{x^2} = -25$

Ans. A, B, C, D

Sol. $g'(x) = f'(x + f(x)) \cdot (1 + f'(x))$

$\implies g'(0) = f'(0)(1 + f'(0)) = 2(1 + 2) = 6.$

$g''(x) = f''(x + f(x)) \cdot (1 + f'(x))^2 + f'(x + f(x)) \cdot f''(x)$

At $x = 0$, $g''(0) = f''(0)(1 + f'(0))^2 + f'(0)f''(0) = (-1)(3)^2 + 2(-1) = -11.$

$\lim\limits_{x\to 0} \dfrac{g(x) - 6x}{x^2} = \lim\limits_{x\to 0} \dfrac{g'(x) - 6}{2x} = \dfrac{g''(0)}{2} = -\dfrac{11}{2};$

Thus, statement (B) is correct.

For statement (C), let $h(x) = f(g(x))$. Note that $h(0) = 0$ and $h'(0) = f'(0)g'(0) = 12.$

Differentiating $h'(x) = f'(g(x))g'(x)$ gives: $h''(x) = f''(g(x))(g'(x))^2 + f'(g(x))g''(x)$

At $x = 0$, $h''(0) = f''(0)(g'(0))^2 + f'(0)g''(0) = (-1)(6)^2 + 2(-11) = -58.$

Applying L'Hôpital's rule twice: $\lim\limits_{x\to 0} \dfrac{f(g(x)) - 12x}{x^2} = \dfrac{h''(0)}{2} = -29$

For statement (D), let $k(x) = g(f(x))$. Note that $k(0) = 0$ and $k'(0) = g'(0)f'(0) = 12.$

Differentiating $k'(x) = g'(f(x))f'(x)$ gives: $k''(x) = g''(f(x))(f'(x))^2 + g'(f(x))f''(x)$

At $x = 0$, $k''(0) = g''(0)(f'(0))^2 + g'(0)f''(0) = (-11)(2)^2 + 6(-1) = -50.$

$\lim\limits_{x\to 0} \dfrac{g(f(x)) - 12x}{x^2} = \dfrac{k''(0)}{2} = -25$

Q7. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function such that $f'(x) = \dfrac{x^2 - 1}{(x^2 + x + 1)\sqrt{x^4 + x^2 + 1}}$ for all $x \in \mathbb{R}$. If $\lim\limits_{x\to\infty} f(x) = 1$, consider the cubic equation $x^3 - 3x + (f(1) - f(-1)) = 0$ Which of the following statements is (are) **TRUE**?

(A) The equation has three distinct real roots.

(B) The product of all the real roots of the equation is $-\dfrac{2}{\sqrt{3}}$.

(C) Exactly one real root of the equation lies in the interval $(1, 2)$.

(D) Exactly one real root of the equation lies in the interval $(-1, 0)$.

Ans. A, C, D

Sol. $f'(x) = \dfrac{1 - \frac{1}{x^2}}{\left(x + 1 + \frac{1}{x}\right)\sqrt{x^2 + 1 + \frac{1}{x^2}}}$

Let $u = x + \dfrac{1}{x} \implies du = \left(1 - \dfrac{1}{x^2}\right)dx.$

$$f(x) = \int \frac{du}{(u+1)\sqrt{u^2-1}}$$

Substituting $u = \sec\theta \implies du = \sec\theta\tan\theta d\theta$:

$$f(x) = \int \frac{\sec\theta}{\sec\theta+1}d\theta = \int \frac{1}{1+\cos\theta}d\theta$$

$$= \tan\left(\frac{\theta}{2}\right) + C = \sqrt{\frac{u-1}{u+1}} + C$$

$$f(x) = \sqrt{\frac{x^2-x+1}{x^2+x+1}} + C$$

Applying the boundary condition $\lim_{x\to\infty} f(x) = 1$: $1 + C = 1 \implies C = 0$

$$\implies f(x) = \sqrt{\frac{x^2-x+1}{x^2+x+1}}$$

$$f(1) = \frac{1}{\sqrt{3}} \quad \text{and} \quad f(-1) = \sqrt{3}$$

$$\implies f(1) - f(-1) = \frac{1}{\sqrt{3}} - \sqrt{3} = -\frac{2}{\sqrt{3}}$$

The cubic equation becomes: $x^3 - 3x - \frac{2}{\sqrt{3}} = 0$. Let $g(x) = x^3 - 3x - \frac{2}{\sqrt{3}}$.

Then $g'(x) = 3(x-1)(x+1)$, giving critical points at $x = \pm 1$.

The local maximum value is $g(-1) = 2 - \frac{2}{\sqrt{3}} > 0$ and the local minimum value is $g(1) = -2 - \frac{2}{\sqrt{3}} < 0$.

Since $g(-1) \cdot g(1) < 0$, the equation possesses three distinct real roots. Hence, statement (A) is true.

The product of the roots is $-\left(-\frac{2}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}$, making statement (B) false.

Evaluating $g(x)$ at specific integer points: $g(-2) = -2 - \frac{2}{\sqrt{3}} < 0$;

$$g(-1) = 2 - \frac{2}{\sqrt{3}} > 0;\ g(0) = -\frac{2}{\sqrt{3}} < 0$$

$$g(1) = -2 - \frac{2}{\sqrt{3}} < 0;\ g(2) = 2 - \frac{2}{\sqrt{3}} > 0$$

Sign changes occur in the intervals $(-2,-1)$, $(-1,0)$, and $(1,2)$.

Thus, exactly one root lies in $(-1,0)$ and exactly one root lies in $(1,2)$.

Q8. Let $\mathcal{C}$ be the coaxal system of circles in the Cartesian plane $\mathbb{R}^2$ such that every member of $\mathcal{C}$ is tangent to the y-axis at the origin $O(0,0)$. Let $\mathcal{L}$ be a family of straight lines parameterized by $m \in \mathbb{R} \setminus \{0\}$, defined by the equation $y = mx + \frac{1}{m}$. For a fixed line $L \in \mathcal{L}$, let C_{λ_1} and C_{λ_2} be the two distinct circles in $\mathcal{C}$ that are tangent to L, with their centers denoted by M_1 and M_2

respectively. Let T_1 and T_2 be the respective points of tangency of L with the circles C_{λ_1} and C_{λ_2}. Which of the following statements is (are) **TRUE**?

(A) The length of the line segment T_1T_2 is equal to the distance between the parallel lines M_1T_1 and M_2T_2, and is given by $\frac{1}{|m|}$.

(B) The circle described with the line segment T_1T_2 as a diameter is orthogonal to every member of the coaxal system $\mathcal{C}$.

(C) The length of the line segment T_1T_2 is equal to the distance between the parallel lines M_1T_1 and M_2T_2, and is given by $\frac{2}{|m|}$.

(D) The area of the right trapezoid formed by the vertices M_1, T_1, T_2, and M_2 is independent of m.

Ans. B, C

Sol. Any circle $C_\lambda \in \mathcal{C}$ has the equation $x^2 + y^2 - 2\lambda x = 0$ with center $M(\lambda, 0)$ and radius $|\lambda|$.

The condition for the line $mx - y + \frac{1}{m} = 0$ to be tangent to C_λ is given by the relation: $\frac{|m\lambda + \frac{1}{m}|}{\sqrt{m^2+1}} = |\lambda|$

Squaring and simplifying yields the quadratic equation $\lambda^2 - 2\lambda - \frac{1}{m^2} = 0$.

The roots λ_1, λ_2 correspond to the centers $M_1(\lambda_1, 0)$ and $M_2(\lambda_2, 0)$, satisfying $\lambda_1 + \lambda_2 = 2$ and $\lambda_1\lambda_2 = -\frac{1}{m^2}$.

The midpoint of M_1M_2 is $(1, 0)$, which is independent of m.

The radical axis of the system $\mathcal{C}$ is the y-axis ($x = 0$), which intersects the line L at $P(0, \frac{1}{m})$.

The power of point P with respect to both circles is $0^2 + (\frac{1}{m})^2 - 2\lambda(0) = \frac{1}{m^2}$

,which implies that the lengths of the tangents are $PT_1 = PT_2 = \frac{1}{|m|}$.

Consequently, $P(0, \frac{1}{m})$ is the midpoint of T_1T_2 and the radius of the circle with diameter T_1T_2 is $\frac{1}{|m|}$.

The equation of this circle is $x^2 + (y - \frac{1}{m})^2 = \frac{1}{m^2} \implies x^2 + y^2 - \frac{2}{m}y = 0$.

Since it is tangent to the x-axis at the origin, it is orthogonal to every member of $\mathcal{C}$, hence (B) is true.

Since M_1T_1 and M_2T_2 are both perpendicular to the tangent line L, they are parallel to each other.

The segment T_1T_2 acts as the perpendicular distance between these two lines, and its length is $PT_1 + PT_2 = \frac{2}{|m|}$, making (C) true.

The area of the right trapezoid $M_1T_1T_2M_2$ is given by:

$$\text{Area} = \frac{1}{2}(R_1 + R_2)(T_1T_2) = \frac{1}{2}|\lambda_1 - \lambda_2| \left(\frac{2}{|m|}\right) = \frac{2\sqrt{m^2+1}}{m^2}$$

This clearly depends on m,

Q9. Consider the following definite integrals:

$$I_1 = \int_0^{\frac{\pi}{3}} \frac{\ln(1+\sqrt{3}\tan x)}{1+\cos 2x + \sqrt{3}\sin 2x}\,dx$$

$$I_2 = \int_0^{\frac{\pi}{3}} \frac{x}{1+\cos 2x + \sqrt{3}\sin 2x}\,dx$$

$$I_3 = \int_0^{\frac{\pi}{3}} \frac{\cos\left(2x - \frac{\pi}{3}\right)\ln(1+\sqrt{3}\tan x)}{1+\cos 2x + \sqrt{3}\sin 2x}\,dx$$

Which of the following statements is (are) **TRUE**?

(A) $I_1 = \dfrac{(\ln 2)^2}{\sqrt{3}}$ (B) $I_2 = \dfrac{\pi \ln 2}{3\sqrt{3}}$ (C) $I_3 = \dfrac{\pi \ln 2}{6} - \dfrac{(\ln 2)^2}{2\sqrt{3}}$

(D) $\displaystyle\int_0^{\frac{\pi}{3}} \frac{\ln(1+\sqrt{3}\tan x) + 3x}{1+\cos 2x + \sqrt{3}\sin 2x}\,dx = \frac{\ln 2}{\sqrt{3}}\left(\ln 2 + \frac{\pi}{2}\right)$

Ans. A, C, D

Sol. $D(x) = 1 + \cos 2x + \sqrt{3}\sin 2x = 1 + 2\cos\left(2x - \dfrac{\pi}{3}\right)$.

Applying the reflection property $\displaystyle\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$ with $a = 0$ and $b = \dfrac{\pi}{3}$

,the transformation $x \to \dfrac{\pi}{3} - x$ yields: $2x - \dfrac{\pi}{3} \to 2\left(\dfrac{\pi}{3} - x\right) - \dfrac{\pi}{3} = -\left(2x - \dfrac{\pi}{3}\right)$

Since $\cos(-\theta) = \cos\theta$, $D(x)$ remains invariant under this transformation.

For the base integral $J = \displaystyle\int_0^{\frac{\pi}{3}} \frac{dx}{D(x)}$, we substitute

$u = \tan x$: $J = \displaystyle\int_0^{\frac{\pi}{3}} \frac{\sec^2 x\,dx}{2(1+\sqrt{3}\tan x)}$

$= \dfrac{1}{2\sqrt{3}}\left[\ln(1+\sqrt{3}\tan x)\right]_0^{\frac{\pi}{3}} = \dfrac{\ln 4}{2\sqrt{3}} = \dfrac{\ln 2}{\sqrt{3}}$

For I_1, the numerator transforms as: $\ln\left(1+\sqrt{3}\tan\left(\dfrac{\pi}{3} - x\right)\right)$

$= \ln\left(1 + \sqrt{3}\cdot\dfrac{\sqrt{3} - \tan x}{1+\sqrt{3}\tan x}\right) = \ln 4 - \ln(1+\sqrt{3}\tan x)$

$I_1 = \displaystyle\int_0^{\frac{\pi}{3}} \frac{\ln 4 - \ln(1+\sqrt{3}\tan x)}{D(x)}\,dx = \ln 4 \cdot J - I_1$

$\Rightarrow 2I_1 = (2\ln 2)\left(\dfrac{\ln 2}{\sqrt{3}}\right) \Rightarrow I_1 = \dfrac{(\ln 2)^2}{\sqrt{3}}$

For I_2, applying the property gives: $I_2 = \displaystyle\int_0^{\frac{\pi}{3}} \frac{\frac{\pi}{3} - x}{D(x)}dx = \frac{\pi}{3}J - I_2$

$\Rightarrow 2I_2 = \dfrac{\pi}{3}\left(\dfrac{\ln 2}{\sqrt{3}}\right) \Rightarrow I_2 = \dfrac{\pi \ln 2}{6\sqrt{3}}$

For I_3, since $\cos\left(2x-\frac{\pi}{3}\right)$ is invariant under the transformation, we have:

$$2I_3 = \ln 4 \int_0^{\frac{\pi}{3}} \frac{\cos\left(2x-\frac{\pi}{3}\right)}{1+2\cos\left(2x-\frac{\pi}{3}\right)}\,dx$$

Substituting $\theta = 2x-\frac{\pi}{3}$ yields:$2I_3 = \frac{2\ln 2}{2}\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\cos\theta}{1+2\cos\theta}\,d\theta$

$$= 2\ln 2 \int_0^{\frac{\pi}{3}} \left(\frac{1}{2} - \frac{1}{2(1+2\cos\theta)}\right)d\theta$$

Evaluating the remaining standard integral using $t=\tan(\theta/2)$:

$$\int_0^{\frac{\pi}{3}} \frac{d\theta}{1+2\cos\theta} = \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{3-t^2} = \frac{1}{\sqrt{3}}\left[\ln\left(\frac{\sqrt{3}+t}{\sqrt{3}-t}\right)\right]_0^{\frac{1}{\sqrt{3}}} = \frac{\ln 2}{\sqrt{3}}$$

$$2I_3 = 2\ln 2\left(\frac{\pi}{6} - \frac{\ln 2}{2\sqrt{3}}\right) \Rightarrow I_3 = \frac{\pi\ln 2}{6} - \frac{(\ln 2)^2}{2\sqrt{3}}$$

$$I_4 = I_1 + 3I_2 = \frac{(\ln 2)^2}{\sqrt{3}} + 3\left(\frac{\pi\ln 2}{6\sqrt{3}}\right) = \frac{\ln 2}{\sqrt{3}}\left(\ln 2 + \frac{\pi}{2}\right)$$

Q10. Match the definite integrals in List-I with their corresponding values in List-II.

List-I	List-II
(P) $\int_0^{\pi/2} \sin^6 x\,dx$	(1) $\frac{\pi}{32}$
(Q) $\int_0^{\pi/2} \sin^4 x\cos^2 x\,dx$	(2) $\frac{5\pi}{32}$
(R) $\int_0^1 x^2\sqrt{1-x^2}\,dx$	(3) $\frac{\pi}{16}$
(S) $\int_0^{\pi/4} \tan^4 x\,dx$	(4) $\frac{\pi}{4}-\frac{2}{3}$
	(5) $\frac{3\pi}{16}$

(A) $(P)\to(2), (Q)\to(1), (R)\to(3), (S)\to(4)$

(B) $(P)\to(1), (Q)\to(2), (R)\to(3), (S)\to(5)$

(C) $(P)\to(2), (Q)\to(3), (R)\to(1), (S)\to(4)$

(D) $(P)\to(5), (Q)\to(1), (R)\to(2), (S)\to(3)$

Ans. A

Sol. **For (P):** Using Walli's formula: $\int_0^{\pi/2} \sin^6 x\,dx = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi}{32}$

For (Q): $\int_0^{\pi/2} \sin^4 x\cos^2 x\,dx = \frac{3\cdot 1\cdot 1}{6\cdot 4\cdot 2}\cdot\frac{\pi}{2} = \frac{\pi}{32}$

For (R): Substitute $x=\sin\theta \implies dx = \cos\theta\,d\theta$.

$$\int_0^1 x^2\sqrt{1-x^2}\,dx = \int_0^{\pi/2} \sin^2\theta\cos^2\theta\,d\theta = \frac{1\cdot 1}{4\cdot 2}\cdot\frac{\pi}{2} = \frac{\pi}{16}$$

For (S): Let $T_n = \displaystyle\int_0^{\pi/4} \tan^n x\,dx$.

$$T_n + T_{n-2} = \frac{1}{n-1}$$

$$T_4 + T_2 = \frac{1}{3}$$

$$T_2 + T_0 = 1 \implies T_2 = 1 - \int_0^{\pi/4} 1\,dx = 1 - \frac{\pi}{4}$$

$$T_4 = \frac{1}{3} - \left(1 - \frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{2}{3}$$

Q11. Match the statements in List-I with their corresponding values in List-II.

List-I	**List-II**
(P) Let C_1 be the circle $x^2+y^2-2x-4y+1=0$. If the image of the center of C_1 with respect to the line $x+y-1=0$ is the center of a circle C_2 which cuts C_1 orthogonally, then the square of the radius of C_2 is equal to	**(1)** 4
(Q) Let the image of the line $x-2y+4=0$ with respect to the line $x-y+1=0$ be the line L. If a circle C_3 with center $(1,6)$ touches the line L, then the square of the radius of C_3 is equal to	**(2)** 5
(R) A circle C_4 passes through the origin and its center lies at the image of the point $(3,4)$ with respect to the line $x=0$. If C_4 cuts the circle $x^2+y^2-2x-2y+k=0$ orthogonally, then the value of k is equal to	**(3)** 2
(S) Let the image of the point $(4,3)$ with respect to the line $x+y-1=0$ be the center of a circle C_7. If C_7 cuts the circle $x^2+y^2=4$ orthogonally, then the square of the radius of C_7 is equal to	**(4)** 9
	(5) 3

(A) $(P)\to(1),(Q)\to(2),(R)\to(3),(S)\to(4)$

(B) $(P)\to(2),(Q)\to(1),(R)\to(4),(S)\to(3)$

(C) $(P)\to(1),(Q)\to(3),(R)\to(2),(S)\to(5)$

(D) $(P)\to(4),(Q)\to(2),(R)\to(1),(S)\to(3)$

Ans. A

Sol. **(P)** The center of C_1 is $O_1(1,2)$ and its radius squared is $r_1^2 = 1^2 + 2^2 - 1 = 4$.

The image of $O_1(1,2)$ with respect to $x + y - 1 = 0$ is $O_2(-1,0)$, which is the center of C_2

.The distance squared between the centers is $O_1O_2^2 = (1+1)^2 + (2-0)^2 = 8$.

For orthogonal circles, $O_1O_2^2 = r_1^2 + r_2^2 \implies 8 = 4 + r_2^2 \implies r_2^2 = 4$. Thus, $(P) \to (1)$.

(Q) The point of intersection of $x - 2y + 4 = 0$ and $x - y + 1 = 0$ is $(2,3)$.

The image of $(0,2)$ on $x - 2y + 4 = 0$ with respect to $x - y + 1 = 0$ is $(1,1)$.

Thus, line L passes through $(2,3)$ and $(1,1)$, which gives its equation as $2x - y - 1 = 0$.

The radius r_3 of C_3 is the perpendicular distance from $(1,6)$ to

L: $r_3 = \dfrac{|2(1) - 6 - 1|}{\sqrt{2^2 + (-1)^2}} = \sqrt{5} \implies r_3^2 = 5$ Thus, $(Q) \to (2)$.

(R) The image of $(3,4)$ with respect to $x = 0$ is $(-3,4)$, which is the center of C_4.

Since C_4 passes through $(0,0)$, its radius squared is $r_4^2 = (-3)^2 + 4^2 = 25$.

For C_5, the center is $(1,1)$ and its radius squared is $r_5^2 = 1^2 + 1^2 - k = 2 - k$.

The square of the distance between their centers is $d^2 = (-3-1)^2 + (4-1)^2 = 25$.

By the condition of orthogonality: $d^2 = r_4^2 + r_5^2 \implies 25 = 25 + 2 - k \implies k = 2$ Thus, $(R) \to (3)$.

(S) The image of $(4,3)$ with respect to $x + y - 1 = 0$ is $(-2,-3)$, which is the center of C_7.

The square of the distance from the origin $(0,0)$ to $(-2,-3)$ is $d^2 = (-2)^2 + (-3)^2 = 13$. For the circle $x^2 + y^2 = 4$, $r_6^2 = 4$.

Applying the condition of orthogonality: $d^2 = r_6^2 + r_7^2 \implies 13 = 4 + r_7^2 \implies r_7^2 = 9$ Thus, $(S) \to (4)$.

Q1. Let $f : [0,1] \to \mathbb{R}$ be a continuous function which is not identically zero on $[0,1]$. Consider a real-valued function $g : \mathbb{R} \to \mathbb{R}$ defined by

$$g(t) = \int_0^1 e^{tx} f(x)\, dx$$

If $g(0) = 0$ and $g'(0) = 0$, then the minimum possible number of distinct roots of the equation $f(x) = 0$ in the open interval $(0,1)$ is equal to ____.

Ans. 2

Sol. Differentiating $g(t)$ under the integral sign and evaluating at $t = 0$ gives:

$$g(0) = \int_0^1 f(x)\, dx = 0 \implies g'(0) = \int_0^1 x f(x)\, dx = 0$$

Since $f(x)$ is continuous and not identically zero, it must take both positive and negative values in $(0,1)$ to satisfy $\displaystyle\int_0^1 f(x)\, dx = 0$.

By the Intermediate Value Theorem, $f(x)$ must have at least one root in $(0,1)$.

Suppose $f(x)$ has exactly one root $c \in (0,1)$ where it changes sign.

Then the function $h(x) = (x-c)f(x)$ does not change sign on $(0,1)$ and is not identically zero, which implies: $\int_0^1 (x-c)f(x)\,dx \neq 0$

$$\int_0^1 (x-c)f(x)dx = \int_0^1 xf(x)dx - c\int_0^1 f(x)dx$$

$$= g'(0) - cg(0) = 0$$

This produces a contradiction. Thus, $f(x)$ must possess at least 2 distinct roots in $(0,1)$.

To verify that 2 is achievable, consider the quadratic polynomial $f(x) = x^2 - x + \frac{1}{6}$.

It satisfies both integral conditions and has exactly 2 distinct roots, $x = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$, both of which lie in $(0,1)$.

Q2. Let $C^\infty(0,\infty)$ denote the vector space of all infinitely differentiable real-valued functions defined on the interval $(0,\infty)$. For a given function $f \in C^\infty(0,\infty)$, a sequence of functions $\{f_n\}_{n=0}^\infty$ is generated recursively according to the following scheme:

$$f_0(x) = f(x)$$

$$f_{n+1}(x) = x\frac{d}{dx}(f_n(x)) + nf_n(x) \text{ for all n} \geq 0 \text{ and x} > 0$$

If the indefinite integral of the function $f_5(x)$ with respect to x is expressed in the form

$$\int f_5(x)\,dx = x\sum_{k=1}^{4} c_k f_k(x) + C$$

where c_k $(k = 1,2,3,4)$ are real constants independent of the choice of f, and C is the arbitrary constant of integration, then the value of c_3 is equal to _____.

Ans. 3

Sol. $\int x^{k-1} f_{n+1}(x)\,dx = \int x^k f_n'(x)\,dx + n\int x^{k-1} f_n(x)\,dx$

Applying integration by parts to the first term on the right-hand side, we choose x^k as the first function and $f_n'(x)$ as the second function: $\int x^k f_n'(x)\,dx = x^k f_n(x) - k\int x^{k-1} f_n(x)\,dx$

$$\int x^{k-1} f_{n+1}(x)\,dx = x^k f_n(x) + (n-k)\int x^{k-1} f_n(x)\,dx$$

For the evaluation of $\int f_5(x)\,dx$, we set $k = 1$ and $n = 4$: $\int f_5(x)\,dx = xf_4(x) + 3\int f_4(x)\,dx$

Applying the reduction relation successively for $n = 3,2,1$ under the condition $k = 1$:

$\int f_4(x)\,dx = xf_3(x) + 2\int f_3(x)\,dx$; $\int f_3(x)\,dx = xf_2(x) + 1\int f_2(x)\,dx$;

$$\int f_2(x)\,dx = xf_1(x) + 0 = xf_1(x)$$

So $\int f_5(x)\,dx = xf_4(x) + 3\left(xf_3(x) + 2\left(xf_2(x) + xf_1(x)\right)\right) + C$

$$\int f_5(x)\,dx = xf_4(x) + 3xf_3(x) + 6xf_2(x) + 6xf_1(x) + C$$

$$\int f_5(x)\,dx = x\left(6f_1(x) + 6f_2(x) + 3f_3(x) + f_4(x)\right) + C$$

$c_3 = 3$

Q3. Consider the function $f : [1, 4] \to \mathbb{R}$ defined by $f(x) = x^3 - 9x^2 + 24x$. Let $c \in (1, 4)$ be the value that satisfies Rolle's theorem for $f(x)$ on the interval $[1, 4]$. If M and m denote the greatest and the least values of $f(x)$ on $[1, 4]$ respectively, then the value of $M - m + c$ is equal to

Ans. 6

Sol. Given $f(x) = x^3 - 9x^2 + 24x$ for $x \in [1, 4]$. At the endpoints, $f(1) = 16$ and $f(4) = 16$. Since $f(1) = f(4)$, Rolle's theorem is applicable.

Differentiating $f(x)$, we get:

$f'(x) = 3x^2 - 18x + 24 = 3(x - 2)(x - 4)$

For $c \in (1, 4)$ such that $f'(c) = 0$, we find $c = 2$. To determine the greatest value M and least value m on $[1, 4]$, evaluate $f(x)$ at the critical points and endpoints:

$f(1) = 16, \quad f(2) = 20, \quad f(4) = 16$

Hence, $M = 20$ and $m = 16$. Thus, $M - m + c = 20 - 16 + 2 = 6$.

Q1. Consider a function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = (a^2 - 1)x^3 + 3(a - 1)x^2 + 6x + 7$ where $a \in \mathbb{R}$. If $f(x)$ is monotonically increasing for all $x \in \mathbb{R}$, then the number of integral values of a in the interval $[-5, 5]$ is

Ans. 8

Sol. For the function $f(x)$ to be monotonically increasing for all $x \in \mathbb{R}$, its derivative $f'(x)$ must be non-negative for all $x \in \mathbb{R}$.

Differentiating $f(x)$ with respect to x: $f'(x) = 3(a^2 - 1)x^2 + 6(a - 1)x + 6 \geq 0$

Dividing the inequality by 3, we get: $(a^2 - 1)x^2 + 2(a - 1)x + 2 \geq 0$

Case 1: If $a^2 - 1 = 0$, then $a = 1$ or $a = -1$.

For $a = 1$, the inequality reduces to $2 \geq 0$, which is always true. Thus, $a = 1$ is a valid value.

For $a = -1$, the inequality reduces to $-4x + 2 \geq 0$, which does not hold true for all $x \in \mathbb{R}$. Thus, $a = -1$ is rejected.

Case 2: If $a^2 - 1 \neq 0$, the quadratic expression is non-negative for all $x \in \mathbb{R}$ if and only if the coefficient of x^2 is positive and the discriminant D is less than or equal to zero.

- Condition 1: $a^2 - 1 > 0 \implies a > 1$ or $a < -1$
- Condition 2: $D = [2(a - 1)]^2 - 4(a^2 - 1)(2) \leq 0 \implies$ (a+3)(a−1)≥0⟹a≤−3 or a≥1

Taking the intersection of Condition 1 and Condition 2 for Case 2 gives: $a \leq -3$ or $a > 1$

Combining the valid solutions from both Case 1 and Case 2, the complete range of a is: $a \in (-\infty, -3] \cup [1, \infty)$

Q2. Let $\lfloor x \rfloor$ and $\{x\}$ denote the greatest integer function and the fractional part function of x, respectively. Let m be the number of points in the interval $(0,2)$ where the function $h(x) = \lfloor x \rfloor + \left| \{x\} - \frac{1}{2} \right|$ is not differentiable. Consider two functions defined in the interval $(1,2)$ as:

$$u(x) = \ln(1 + \{x\}) \quad \text{and} \quad v(x) = \frac{1}{1 - \{x\}}$$

If d is the value of the derivative of $u(x)$ with respect to $v(x)$ at $x = \frac{3}{2}$, then the value of $\frac{m}{d}$ is equal to

Ans. 18

Sol. To find m, analyze the differentiability of $h(x)$ on the interval $(0,2)$:

For $x \in (0,1)$, $\lfloor x \rfloor = 0$ and $\{x\} = x$, so $h(x) = \left| x - \frac{1}{2} \right|$,

which is non-differentiable at the corner point $x = \frac{1}{2}$.

For $x = 1$, $\lim\limits_{x \to 1^-} h(x) = 0 + \left| 1 - \frac{1}{2} \right| = \frac{1}{2}$ and $\lim\limits_{x \to 1^+} h(x) = 1 + \left| 0 - \frac{1}{2} \right| = \frac{3}{2}$.
Since $h(x)$ is discontinuous at $x = 1$, it is non-differentiable at $x = 1$.

For $x \in (1,2)$, $\lfloor x \rfloor = 1$ and $\{x\} = x - 1$, so $h(x) = 1 + \left| x - \frac{3}{2} \right|$,

which is non-differentiable at the corner point $x = \frac{3}{2}$.

Thus, the points of non-differentiability in $(0,2)$ are $x = \frac{1}{2}, 1, \frac{3}{2}$, giving $m = 3$.

To find d, consider $x \in (1,2)$ where $\{x\} = x - 1$:

$u(x) = \ln(1 + x - 1) = \ln x \implies \dfrac{du}{dx} = \dfrac{1}{x}$

$v(x) = \dfrac{1}{1 - (x-1)} = \dfrac{1}{2 - x} \implies \dfrac{dv}{dx} = \dfrac{1}{(2-x)^2}$

The derivative of $u(x)$ with respect to $v(x)$ is: $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx} = \dfrac{(2-x)^2}{x}$

$d = \dfrac{\left(2 - \frac{3}{2}\right)^2}{\frac{3}{2}} = \dfrac{\frac{1}{4}}{\frac{3}{2}} = \dfrac{1}{6}$

$\dfrac{m}{d} = \dfrac{3}{\frac{1}{6}} = 18$

Q3. Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function representing a curve $y = f(x)$. Let T and N denote the length of the tangent and the length of the normal to the curve at any point $P(x, y)$ on it, respectively. Suppose that $f'(x) > 0$ for all $x > 0$ and the function satisfies the relation

$$\frac{1}{y}\left(\frac{N}{T}\right) = x + \frac{1}{x}$$

for all $x > 0$. If $f(2) = 2e^2$, then the value of $\ln\left(\dfrac{f(4)}{4}\right)$ is equal to

Ans. 8

Sol. For a curve $y = f(x)$, the length of the normal N and the length of the tangent T at any point $P(x, y)$ are defined as:

$$N = \left| y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} \right|; T = \left| \frac{y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}{\frac{dy}{dx}} \right| \implies \frac{N}{T} = \frac{dy}{dx}$$

$$\frac{1}{y}\frac{dy}{dx} = x + \frac{1}{x}$$

Integrating both sides with respect to x: $\int \frac{1}{y} dy = \int \left(x + \frac{1}{x}\right) dx \implies \ln y = \frac{x^2}{2} + \ln x + C$

Applying the initial condition $f(2) = 2e^2$: $\ln(2e^2) = \frac{2^2}{2} + \ln 2 + C \implies$ $\ln 2 + 2 = 2 + \ln 2 + C \implies C = 0$

Therefore, the general expression for the curve is: $\ln y = \frac{x^2}{2} + \ln x$

$$\ln(f(4)) = \frac{4^2}{2} + \ln 4 = 8 + \ln 4$$

$$\ln\left(\frac{f(4)}{4}\right) = \ln(f(4)) - \ln 4 = (8 + \ln 4) - \ln 4 = 8$$